MIL-F-50730(MU)
28 April 1972
SUPERSEDING
FA-PD-ML-2726, Rev. 0
21 March 1967

MILITARY SPECIFICATION

FUZE, MTSQ, M564E2,LOADING AND PACKING

1. SCOPE

1.1 This specification covers Fuze, MTSQ, M564E2 Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATION

Military

MIL-P-60412	- Packaging, Packing and Marking for Shipment of Artillery Type and Rocket Fuzes: General Specification for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-331	- Fuze and Fuze Components Environmental and Performance Test for
MIL-STD-1167	- Ammunition Data Card

DRAWINGS

Munitions Command

B8595510	- Booster Pellet
F10520700	- Fuze, MTSQ, M564E2 Assembly
IEL-10520700	- Inspection Equipment List for Fuze, M564E2 Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

FSC 1390

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3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, references standards and specificiations.

3.2 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.2.1 Foreign matter.-No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

3.3 Waterproofness.-The fuze shall comply with MIL-STD-331, Test No. 108, except that the duration of immersion shall be five (5) minutes and the pressure shall be three (3) psi gauge.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Lot formation.-Each inspection lot of fuzes shall be comprised of components (except explosive components) each produced by one manufacturer, by one process, in accordance with the same drawing and revision and specification and revisions, as applicable, and assembled on one assembly line. Each explosive component within a lot of assembled fuzes shall be limited to one manufacturer and one interfix number.

4.2 First article sample.

4.2.1 Inspection provisions for first article not required.

4.3 Inspection provisions.

4.3.1 Submission of product.-The product shall be presented in accordance with MIL-STD-105.

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4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major defects shall be performed on a single defect bases in accordance with MIL-STD-105. For each classification of defects contained herein, the AQL given shall be applied to each major defect contained herein. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination.

4.3.2.1 Fuze subassembly Drawing F10520700 should be visually examined for the following in order to determine condition of material as received:

	a.	Fuze not set on safe "S"	(Critical)
*	b.	Safety adapter rotor armed or partially armed as viewed through the flash hole (Detonator or rotor gear rivet visible)	(Critical)
*	c.	Foreign matter in flash hole	(Major - AQL = 0.40%)
	d.	Fuze damaged	(Major - AQL = 0.40%)

* (to be checked after removing dust seal)

4.3.2.2 Closing Cup Assembly, Drawing B8595506, covering a detail of Drawing F1020700.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR AQL = 0.40%	
101. Explosive above flush	Gage
102. Explosive below flush exceeds max.	Gage
103. Cup damaged	Visual

4.3.2.3 Booster Pellet Drawing B8595510, covering a detail of Drawing F10520700.

CRITICAL	
None defined	
MAJOR AQL = .40%	
101. Diameter, improper	Gage
102. Width, improper	Gage
103. Weight, improper	Gage

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4.3.2.4 Fuze, MTSQ, M564E2 Assembly, Drawing F10520700.

CRITICAL		
1.	Explosive material on internal thread of booster cup (small amount of explosive dust is acceptable)	Visual
2.	Fuze not set on safet "S"	Visual
MAJOR	AQL = 0.40%	
101.	Length from flange to bottom of cup exceeds maximum	Gage
102.	Booster cup split or damaged	Visual
103.	Booster pellet missing (as indicated by booster cup contacting shoulder of body)	Visual
104.	Booster pellet moves when assembly is shaken	Visual

4.3.2.5 Packaging, packing and marking.-The sample size and acceptance criteria for packaging, packing and marking shall be in accordance with MIL-P-60412.

4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Binder-lubricant content of tetryl.-A minimum of one sample (8 ounces) to a maximum of five (5) samples (40 ounces) shall be required from each lot to determine the content of binder-lubricant of the tetryl in each lot of boosters. Failure of the sample to meet the requirements specified on Drawing B8595510 shall be cause for rejection of the lot.

4.3.3.2 Process control audit.-Periodic unscheduled monitoring shall be conducted for the process operations on the following assemblies to determine compliance with the requirements of the applicable drawings.

a. Fuze Subassembly, Drawing F10520701

(1) Removal of dust cover and insertion of closing cup assembly.

b. Fuze, MTSQ, M564E2 Assembly, Drawing F10520700.

(1) Insertion of booster pellet in closing cup.

(2) Application of sealing compound to cup threads.

(3) Tightening of booster cup on safety adapter.

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4.3.3.3 Waterproofness test.-Four fuze assemblies per F10520700 shall be selected from each lot and subjected to the waterproofness test. Evidence of water entry through the booster cup or the booster cup joint shall be cause for rejection of the lot. Water entry into the fuze (O-ring, etc.) shall not be cause for rejection of the lot. Record and forward data to technical agency.

4.3.4 Inspection equipment.-The examinations and tests shall be made using equipment listed in IEL-10520700. Equipment design shall be in accordance with the code designation on the applicable IEL.

4.4 Test methods and procedures.

4.4.1 Binder-lubricant content of tetryl.-A sample of approximately 8 ounces of tetryl shall be selected and determined as follows: An accurately weighed portion of five (5) grams of the sample shall be transferred to a tared medium porosity sintered glass filtering crucible. The sample shall be extracted with four 10-milliliter portions of acetone. Each portion shall be maintained in contact with the residue for approximately five (5) minutes before applying suction. Any lumps which may persist shall be broken up. The side of the crucible shall be washed down with each 10 milliliter portion of acetone to insure complete removal of the tetryl. The residue shall be aspirated until the odor of acetone is no longer detectable. The crucible and contents shall be placed in a drying oven at 100 degrees (°) to 105° Centigrade. (C) for one-half hour, cooled in a desiccator and weighed. The above extraction, drying and weighing procedure shall be repeated with individual 10 milliliter portions of acetone until the loss in weight on subsequent extractions totals less than 0.005 gram. The weight of the residue in the crucible shall be calculated to the precentage of total binder-lubricant in the sample to determine compliance with the requirements specified on the applicable drawing.

4.4.2 Waterproofness.-The test shall be performed in accordance with the procedure specified in MIL-STD-331, Test No. 108, except as provided in 3.3. The fuze shall be disassembled and inspected as follows:

a. Remove booster cup, safety adapter, head, lower cap, check all visible internal surfaces for evidence of water entry.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-Packaging, packing, and marking shall be as specified in the contract in accordance with MIL-P-60412.

5.2 Data cards.-Data card information shall be as specified in MIL-STD-1167.

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6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

Custodian:

Army - MU

Preparing activity:

Army - MU(FA)

Project No. 1390-A213

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